Department of Computer Science and Engineering
UIET, CSJM University, Kanpur

Unix Laboratory Assignment–2

Inspecting, Searching and Controlling a Unix System

Dr. Sandesh Gupta

Purpose of This Lab

In the previous lab, you created, copied, moved and organized files and directories. In this lab, you will learn to inspect file contents, search for information, connect commands, understand permissions and observe running processes.

Commands and Symbols Used in This Lab

`ls -l`, `ls -a`, `ls -lh`	Detailed listings, hidden names and readable file sizes.
`cat`, `less`, `head`, `tail`, `wc`	Display or count the contents of text files.
`grep`, `find`	Search inside files or search for file and directory names.
`rm -i`, `rmdir`	Safely remove a file or an empty directory.
`chmod`, `ls -l`	Change and inspect permissions.
`ps`, `top`, `sleep`, `jobs`, `kill`	Observe and control processes.
`man`, `-help`, `apropos`, `type`	Obtain help about commands.
`*`, `?`, `>`, `»`, `|`	Wildcards, redirection and pipes.

Safety Rule

Perform the entire lab inside `~/unix-lab-02`. Before removing anything, use `pwd` and `ls` to confirm the location and target. Do not use `rm -r`, `sudo` or `chmod 777`.

Preparation: Build a Safe Practice Area

1. Go to your home directory and create `unix-lab-02`.
2. Inside it, create `notes`, `programs`, `reports` and `temporary`.
3. Create these empty files: `notes/unix.txt`, `notes/commands.txt`, `programs/hello.c`, `programs/sum.c`, `programs/test1.c`, `programs/test2.c`, `reports/week1.txt` and `temporary/old.txt`.
4. Use `pwd` and `ls -R` to verify the workspace.

Problem 1: Become a File Inspector

Situation

You are preparing the UIET C-programming workspace. Before using a file, you must be able to identify it, inspect it and locate it quickly.

1. Compare the output of `ls`, `ls -l`, `ls -a` and `ls -lh`.
2. Use `echo` with `>` to write `pwd shows location` into `notes/commands.txt`. Then use `echo` with `»` to append `ls shows names` and `cd changes directory`.
3. Display the file using `cat`; then open it using `less`. Press `q` to leave `less`.
4. Use `head` and `tail` on the file. Try the option `-n 2` with both commands.
5. Use `wc -l` to count its lines.
6. Search for the word `shows` using `grep`.
7. Repeat the search with `grep -n` and observe the difference.
8. From the main practice directory, use `find` to locate:

© 2026 Sandesh Gupta

- every file ending in `.c`; and
- the file named `commands.txt`.

Be ready to explain: when `less` is more useful than `cat`, and what `-n` adds to the output of `grep`.

Problem 2: Use Patterns, Redirection and Pipes

Situation

The project now contains several files. Instead of handling every name separately, use patterns and connect small Unix commands.

A. Wildcards

1. Use `ls programs/*.c` to list every C source filename.
2. Use `ls programs/test?.c`. Identify what the `?` matches.
3. Use a suitable `*` pattern to list every `.txt` file inside `notes`.
4. Always preview a wildcard with `ls` before using it with another command.

B. Redirection

1. Save the current date and time in `reports/system-report.txt` using `>`.
2. Append your current directory to the same file using `>>`.
3. Append a long listing of the `programs` directory.
4. Display the completed report using `cat`.
5. Explain why using `>` again would be different from using `>>`.

C. Pipes

1. Send the output of `ls -l` to `less`.
2. Use `find` to locate all `.c` files, then pipe the result to `wc -l`.
3. Pipe `cat notes/commands.txt` into `grep shows`.
4. Explain what information travels through the `|` symbol.

Problem 3: Permissions, Processes and Help

Situation

A Unix user must know who can access a file, which programs are running, and where to find reliable help.

A. Permissions

1. Create an empty file named `programs/demo.sh`.
2. Use `ls -l programs/demo.sh` and identify the permission characters.
3. Add execute permission for the owner using `chmod u+x programs/demo.sh`.
4. Use `ls -l` again and identify what changed.
5. Remove the owner's execute permission using `chmod u-x programs/demo.sh`; verify the result.

B. Processes

1. Use `ps` to display your processes.
2. Start `sleep 120 &` and observe that the prompt returns.
3. Use `jobs` and `ps` to find the new process.

© 2026 SANDESH GUPTA UIET CSJM UNIVERSITY, KANPUR

4. Stop it using **kill** with the correct process ID.
5. Start **top**, observe the display briefly, and press q to leave.

C. Help

1. Open **man ls**; search for **-a** using **/-a**; press q to leave.
2. Compare **man cp** with **cp -help**.
3. Use **apropos "copy files"**.
4. Run **type cd** and **type ls**. Observe whether the shell describes them in the same way.

Safe Cleanup Practice

1. Use **pwd** and confirm that you are inside **unix-lab-02**.
2. List **temporary** and confirm that it contains only **old.txt**.
3. Remove the file using **rm -i temporary/old.txt**.
4. Remove the now-empty directory using **rmdir temporary**.
5. Verify the result using **ls**.

Final Demonstration

Before leaving the lab, show the instructor:

- **reports/system-report.txt**;
- the result of finding and counting all **.c** files;
- the permission change on **programs/demo.sh**;
- one completed **man** search; and
- the safely removed **temporary** directory.

CSJM UNIVERSITY, KANPUR

© 2026 Sandesh Gupta